

Density Ridge Selective Prediction for LLM and VLM Hallucination Detection under Calibration-Label Scarcity

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Abstract—Hallucination detection in large language and vision-language models is increasingly framed as selective prediction, where a detector assigns a confidence score and abstains when confidence is low. Unsupervised sampling detectors (Semantic Entropy, EigenScore) avoid labels but plateau in quality, while supervised probes (SAPLMA) attain stronger in-distribution scores yet degrade sharply when calibration labels are scarce. We recover the response manifold of an LLM as the density ridge of a kernel density estimate built on a six-dimensional kinematic feature map of hidden state generation trajectories. A test generation is scored by the negated Euclidean distance from its projected feature point to the nearest ridge vertex, yielding a low-dimensional geometric skeleton of the stochastic output distribution. We evaluate against Semantic Entropy, SAR, EigenScore, SAPLMA, and log-probability on seven QA benchmarks (HaluEval-QA, TriviaQA, GSM8K, POPE, ScienceQA, A-OKVQA) using nine text and vision LLMs in a deliberately label-scarce protocol ($n_{\text{cal}}=200$ queries, $N=5$ generations). Our ridge-based score beats on AUROC with 5-20 points gain, while demonstrating tempered degradation under calibration-label scarcity.

Index Terms—hallucination detection, selective prediction, density ridge, kernel density estimation, large language models, uncertainty quantification.

I. INTRODUCTION

Selective prediction abstains when a confidence score falls below a threshold. Two dominant families dominate the literature on LLM hallucination detection: sampling-based unsupervised detectors that probe output dispersion [1], [2], and supervised hidden state probes that train a classifier on labeled correctness annotations. The latter outperform the former under abundant labels but suffer in deployments where calibration data is scarce [3]. We exploit a complementary signal: the geometry of generation-time hidden state trajectories. Repeated sampling at a fixed query traces a multimodal distribution in embedding space whose modes encode distinct response strategies. Recent work on the curvature evolution of LLM trajectories has shown such kinematics to be diagnostic of reasoning quality [11], [12]. We characterize this distribution by the *density ridge* [4], [5] of a KDE fitted to kinematic features of correct trajectories, and score test generations by proximity to this 1-D ridge.

Contributions.: (i) A response-manifold detector recovering the LLM response manifold as a density ridge over supervised, low-dimensional trajectory-curvature features. (ii)

A label-scarce evaluation across nine models and seven benchmarks against Semantic Entropy, log-probability, and topology baselines. (iii) Ablations isolating the contribution of ridge geometry across three parameterization variants and kernel configurations.

II. METHOD

Kinematic feature map. Given training queries $\mathcal{Q}_{\text{train}} = \{q_i\}$ with correctness labels $y_i \in \{0, 1\}$, each sampled completion j induces a hidden-state trajectory $\mathbf{H}_{i,j} \in \mathbb{R}^{T_{i,j} \times D}$ of final-layer, last-token states. With $\Delta \mathbf{h}_t = \mathbf{h}_{t+1} - \mathbf{h}_t$ and discrete curvature $\kappa_t = \|\Delta \mathbf{h}_{t+1} - \Delta \mathbf{h}_t\|_2 / [\frac{1}{2}(\|\Delta \mathbf{h}_t\|_2 + \|\Delta \mathbf{h}_{t+1}\|_2)]^2$, the feature map $\varphi : \mathbb{R}^{T \times D} \rightarrow \mathbb{R}^6$ collects $(K_n, R^2, K_{\text{max}}, \|\Delta \mathbf{h}\|, \|\mathbf{v}\|_{\text{max}}, \tau^*)$: mean and peak curvature, displacement linearity, mean and peak per-step displacement, and the normalized argmax of κ_t [11]. Each generation becomes $\mathbf{x}_m = \varphi(\mathbf{H}_m) \in \mathbb{R}^6$, z -scored by training statistics $\mu_{\text{train}}, \sigma_{\text{train}}$.

Ridge construction. On the correct subset $\tilde{\mathcal{X}}^+ = \{\tilde{\mathbf{x}}_m : y_{\sigma(m)}=1\}$ we fit a Gaussian KDE $\hat{p}$ with bandwidth via Scott’s rule in $d=6$. With $\mathbf{g} = \nabla \log \hat{p}$ and Hessian $\nabla^2 \log \hat{p} = \mathbf{V} \mathbf{A} \mathbf{V}^\top$ ($\lambda_1 \leq \dots \leq \lambda_6$), the 1-D density ridge [4], [5] is $\mathcal{R}_1 = \{\tilde{\mathbf{x}} : \mathbf{V}_\perp^\top \mathbf{g} = \mathbf{0}, \lambda_5 < 0\}$, with $\mathbf{V}_\perp \in \mathbb{R}^{6 \times 5}$ spanning the normal subspace. SCMS iterates the projected gradient $\mathbf{V}_\perp \mathbf{V}_\perp^\top \mathbf{g}$ to fixed points $\mathcal{V} = \{\mathbf{v}_k\}$. Intrinsic-dimension verification via TwoNN [7] supports the $r=1$ clamp. Three chart variants (Table I) furnish usable coordinates: global LTSA, the Hastie–Stuetzle geodesic arclength [8], and a stitched local-chart atlas [9], [10].

Score and OOD interpretation. For a test query with trajectories $\{\mathbf{H}_{\star,j}\}$, the off-ridge distance is $z_{\text{off}}(\tilde{\mathbf{x}}_{\star,j}) = \min_k \|\tilde{\mathbf{x}}_{\star,j} - \mathbf{v}_k\|_2$ and the score $s(q_\star) = -\frac{1}{N_\star} \sum_j z_{\text{off}}(\tilde{\mathbf{x}}_{\star,j})$. Because $\mathcal{R}_1$ is fit on ν^+ alone, incorrect points are OOD regardless of their own density. Standard regularity [6] gives $d_H(\hat{\mathcal{R}}_1, \mathcal{R}_1) = O(h^2) + O_{\mathbb{P}}(\sqrt{\log n / (nh^8)})$ and an expected score gap $\mathbb{E}_{\nu^-}[z_{\text{off}}] - \mathbb{E}_{\nu^+}[z_{\text{off}}] \geq \Delta - \rho^+ - O(h^2)$ concentrating at rate $N_\star^{-1/2}$.

III. EXPERIMENTS

A. Setup

We evaluate eight text and vision-language models, including Mistral-7B-Instruct-v0.3, Gemma-2-9B-IT, LLaVA-

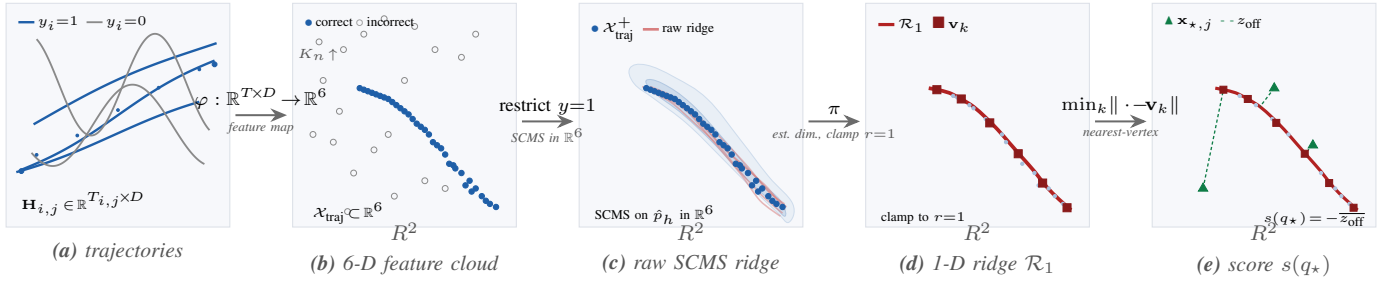


Fig. 1: Trajectory-branch SCMS confidence pipeline. **(a)** For each query q_i and each of N generations, the hidden-state trajectory $\mathbf{H}_{i,j} \in \mathbb{R}^{T_i, j \times D}$ is colored by the binary correctness label y_i . **(b)** The feature map $\varphi(\mathbf{H}) = (K_n, R^2, \dots)$ projects each trajectory to a point $\mathbf{x}_m \in \mathbb{R}^6$, producing $\mathcal{X}_{\text{traj}}$ (two coordinates shown). **(c)** The KDE $\hat{p}_h$ over the trusted subset $\mathcal{X}_{\text{traj}}^+ = \{\mathbf{x}_m : y_{\sigma(m)} = 1\}$ is run through SCMS to extract the raw ridge. **(d)** Intrinsic-dimension verification clamps $r=1$, yielding the 1-D ridge $\mathcal{R}_1$ sampled as vertices $\mathcal{V} = \{\mathbf{v}_k\}_{k=1}^K$. **(e)** A test query q_* contributes trajectory points $\mathbf{x}_{*,j}$; the score is the negated mean perpendicular distance to the nearest ridge vertex.

TABLE I: Ridge embedding variants. All three start from the same SCMS density-ridge estimate and differ in how the ridge is converted into coordinates.

Method	Constructs	Output Dimension	Captures
Ridge LTSA Chart	1 global LTSA	r	Global tangent structure
Ridge Arclength	1-D geodesic on ridge graph	$r+1$ (arclen + z_{off})	Progress along a curve
Ridge Atlas	Many local charts stitched	$r+1$ (local + z_{off})	Curved / multi-patch manifolds

1.5-7B, Idefics3-8B-Llama3, and SmolVLM-Instruct, on six QA benchmarks spanning textual factuality (HaluEval-QA, TriviaQA), mathematical reasoning (GSM8K), and multimodal grounding (POPE, ScienceQA, A-OKVQA). A *cell* denotes one (model, dataset, quantifaction) combination. To simulate data scarce deployments, queries are retained only if they yield at least $\geq 3 - 6$ correct generations, and calibrations uses $n_{\text{cal}} = 200$ queries with $N = 5$ generations each (test size is 60). We report AUROC and PRR (higher is better), and AURC and AUGRC (lower is better). The head-to-head comparison is restricted to cells common to all detectors.

B. Baselines

We compare against unsupervised sampling detectors Semantic Entropy [1], log-probability, and topology-based metrics such as topological LID-MLE and persistent-homology, the TRACED mean curvature scalar [11], and naive embedding-geometry baselines ($k\text{NN}-R^2$, PCA-based, flow matching to the correct-class centroid). Recent representation-manifold approaches have been explored elsewhere for safety [13]; the multi-dimensional nature of LLM features [14] motivates the 6-D kinematic descriptor over scalar summaries. The ordering of detectors by AUROC is: ridge > log-probability > single-scalar trajectory summaries > naive geometry > topological summaries. Negative-control scalars (initial-state distance, weight-norm) and maximum-token log-probability invert to PRR < 0, as anti-correlated signals should.

C. Ablations

a) Kernel configuration.: Sweeping eleven SCMS kernel variants together with three naive-geometry baselines (PC1,

$k\text{NN}$, Mahalanobis) within each cell, the canonical kernel (fixed bandwidth via Scott’s rule, uniform weights, sample covariance, $r=1$ constraint) attains the best mean rank (3.27), and every SCMS variant outranks all three naive baselines (mean ranks ≥ 11). The ridge structure, not generic embedding distance, is what separates correct from hallucinated generations.

D. Main Results

Table II reports per-cell selective prediction metrics for three representative detector classes: the ridge score (Ridge Arclength, Ridge LTSA Chart, or Ridge Atlas), a non-ridge geometric or topological baseline (Topology LID-MLE, $k\text{NN}-R^2$, Flow Matching, Semantic Entropy, or TRACED mean curvature), and calibrated sequence log-probability. The ridge score is the most performant predictor on every metric for all cells.

Quantification trials across detectors and datasets. The log-probability baseline is the strongest non-ridge competitor on the textual factuality and grounding splits: it attains AUROC 0.913 on Mistral/HaluEval-QA, 0.817 on LLaVA/POPE, and 0.79–0.80 on the SmolVLM and Idefics3 multimodal cells. Yet on every cell the ridge variant improves AUROC by 5–20 points absolute and concurrently reduces AURC and AUGRC. The largest gains occur on the multimodal grounding datasets (A-OKVQA, ScienceQA), where Idefics3 sees AUROC rise from 0.795 (logP) to 0.972 (Ridge LTSA Chart) and SmolVLM/ScienceQA rises from 0.790 to 0.990 (Ridge Arclength), with AUGRC roughly halved. The non-ridge baselines are markedly less stable: Topology LID-MLE degrades to near-chance (AUROC 0.470–0.511) on SmolVLM/A-OKVQA

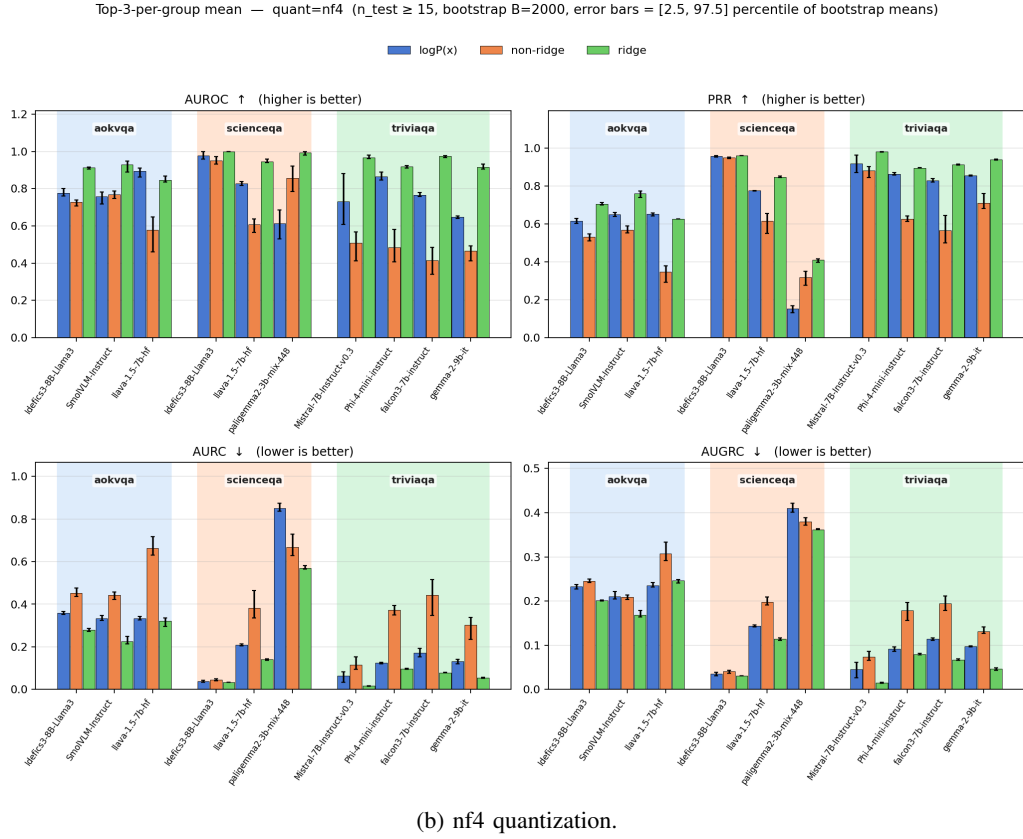
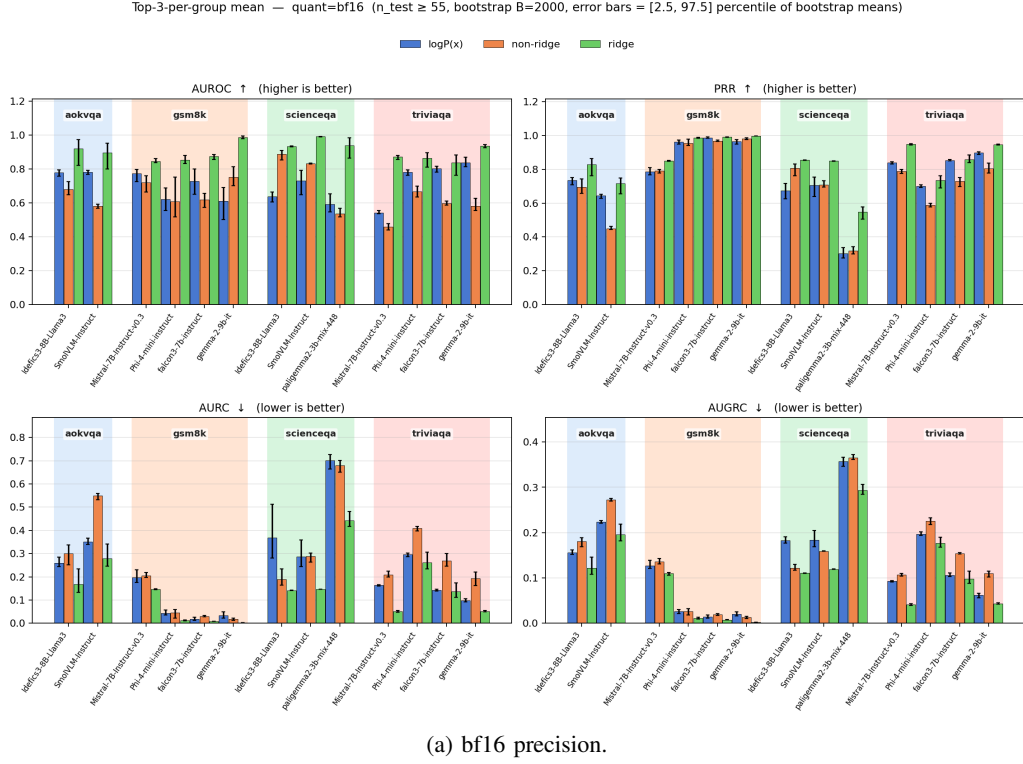


Fig. 2: Top-3 detectors across (model, dataset) cells, evaluated under bf16 precision (top) and nf4 quantization (bottom). Panels show AUROC and PRR (higher is better) and AURC and AUGRC (lower is better).

TABLE II: Selective prediction comparison on select methods ($q = 200, N = 5, \text{test size} = 60$). Ridge estimation methods used on hidden state trajectory sequences are compared against calibrated log probability, and a non-ridge method. Arrows $\uparrow/\downarrow$ indicate higher/lower is better. Most performant predictor per (model, dataset) is in **bold**.

Model	Dataset	Method (Variant/Scorer)	AUROC $\uparrow$	AURC $\downarrow$	PRR $\uparrow$	AUGRC $\downarrow$
Idefics3-8B-Llama3	A-OKVQA	logP(x) (<i>Sequence logP(x)</i>)	0.795	0.244	0.752	0.152
		Topology LID-MLE (<i>Neg LID-MLE</i>)	0.725	0.252	0.744	0.169
		Ridge LTSA Chart Baseline (<i>PR-dim/Ridge Proximity</i>)	0.972	0.132	0.864	0.108
SmolVLM-Instruct	A-OKVQA	Baseline logP(x) (<i>Sequence logP(x)</i>)	0.791	0.344	0.651	0.221
		Topology LID-MLE (<i>Neg LID-MLE</i>)	0.470	0.593	0.402	0.299
		Ridge LTSA Chart (<i>PR-dim/Ridge Proximity</i>)	0.951	0.245	0.750	0.182
Idefics3-8B-Llama3	ScienceQA	Baseline kNN (<i>kNN-R² proximity</i>)	0.907	0.163	0.833	0.117
		Topology LID-MLE (<i>Neg LID-MLE</i>)	0.511	0.526	0.479	0.214
		Ridge ArcLength (<i>Spherized/Ridge Proximity</i>)	0.934	0.140	0.856	0.110
SmolVLM-Instruct	ScienceQA	Baseline logP(x) (<i>Sequence logP(x)</i>)	0.790	0.243	0.753	0.169
		Flow Matching (<i>Neg dist. to correct centroid</i>)	0.769	0.326	0.670	0.175
		Ridge ArcLength (<i>PR-dim/Ridge Proximity</i>)	0.990	0.146	0.850	0.119
Mistral-7B-Instruct-v0.3	HaluEval-QA	Baseline logP(x) (<i>Sequence logP(x)</i>)	0.913	0.202	0.794	0.147
		Traced Mean Curvature	0.892	0.203	0.793	0.152
		Ridge ArcLength (<i>Shrinkage/Ridge Proximity</i>)	0.971	0.169	0.826	0.132
LLaVA-v1.5-7B	POPE	Baseline logP(x) (<i>Mean logP(x) per token</i>)	0.817	0.106	0.891	0.077
		Semantic Entropy	0.740	0.133	0.864	0.093
		Ridge LTSA Chart (<i>TRiE/Ridge Proximity</i>)	1.000	0.044	0.953	0.040
Gemma-2-9B-IT	GSM8K	Baseline kNN (<i>kNN-R² proximity</i>)	0.813	0.012	0.988	0.010
		Neg. H ₀ total persistence	0.637	0.024	0.975	0.018
		Ridge Atlas (<i>TRiE/Ridge Proximity</i>)	0.994	0.002	0.998	0.002

Method legend. All methods, except the log probability baselines, produce a per-query scalar confidence score, where higher confidence can be used as a selective prediction signal; scores are negated to standardized orientation to higher being more confident. Trajectories are acquired from sequences of hidden state final layer last tokens across T autoregressive decoding steps (shape (T, D)), aggregated across N sampled generations. *Ridge-based (ours): Ridge ArcLength, Ridge LTSA Chart, Ridge Atlas*, described in detail in Table I, fit a subspace constrained mean shift (SCMS) principal ridge on a projection of the hidden states of the training set (correct-only subset); the score is the perpendicular distance z_{off} from the test query’s hidden state to its nearest ridge vertex (*Ridge Proximity*). Ablated variants of SCMS variants are utilized to ascertain the effect of ridge projection and density estimation on trajectory-computed selective prediction; ablation variants in the table include *Shrinkage* (covariance ablation), *Trajectory Ridge Estimate (TRiE)* (no ablation), *Spherized* (projection ablation via per-row L_2 normalization instead of identity projection), and *PR-dim* (projection and covariance ablation). *Baselines: Baseline logP(x)* — *Sequence*: sum of token log-probs; *Mean/token*: length-normalized. **Baseline kNN ($k\text{NN-}R^2$ proximity)** — local R^2 -style statistic over the labeled training set near the query’s hidden state. **Semantic Entropy** [1] — NLI-clusters N generations into semantic equivalence classes, Shannon entropy over cluster probabilities (negated for confidence). **Flow Matching** — train a CFM vector field on correct-class hidden states (PCA to TwoNN dim); integrate ODE backward to obtain the base-Gaussian latent z ; score = negative distance from z to correct-class centroid. **Topology LID-MLE** — negated MLE (maximum likelihood estimate) of local intrinsic dimension (LID) at the query; query’s local hidden-state neighborhood. **Traced Mean Curvature** — trace of the mean-curvature tensor along each (T, D) generation trajectory, aggregated across the N generations [11]. **Neg. H₀ total persistence** — sum of bar lengths in the H₀ persistence diagram of the per-query hidden state under a Vietoris–Rips filtration (negated).

and Idefics3/ScienceQA, while Flow Matching to the correct-class centroid (AUROC 0.769 on SmolVLM/ScienceQA) and Semantic Entropy (0.740 on LLaVA/POPE) underperform log-probability on the cells where they are the strongest non-ridge entrant. On the reasoning-heavy text cells where log-probability is already strong (HaluEval-QA), the ridge variant still recovers a 4–6 point AUROC margin while trimming AURC by roughly 15%, indicating that the geometric signal is complementary to, not redundant with, token-level confidence. Figure 2 aggregates the top-3 detectors per cell under bf16 (top) and nf4 (bottom) precision, illustrating that the ridge–logP–non-ridge ordering persists under aggressive quantization.

IV. DISCUSSION AND CONCLUSION

That the ridge detector exceeds both unsupervised sampling detectors and sequence log-probability under deliberately scarce calibration labels indicates that what separates faithful from hallucinated generations is the *shape* of the trajectory-feature space, not its raw location. Every SCMS variant outranks PC1, $k\text{NN}$, and Mahalanobis: generic embedding distance is insufficient, and the recovered 1-D manifold density ridge is what is utilized by the score. **Limitations.** (i) The head-to-head comparison is restricted to the cells common to all detectors. (ii) Semantic Entropy is at chance on closed-form slices, which is an expected degeneracy under data and label scarcity. (iii) The supervised projection front-end requires both-class labels at fit time. Natural extensions include genuine per-axis kernel ablations and lifting distribution-free conformal

guarantees [15], [16] from a companion probe onto the ridge score itself.

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